Project Proposal

PROJECT NAME	Internal Assistant for Enterprise Knowledge (Slack App Bot)		
PROJECT LOCATION	Swinburne University of Technology		
EST. START DATE	Sep 29th, 2026	**EST. FINISH DATE**	Dec 6th, 2026
PROJECT LEADER	Mr. Truong Ngoc Gia Hieu	**UNIVERSITY**	Swinburne University of Technology
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SUMMARY	The project focuses on designing and developing an official Slack app bot at the enterprise level to solve the problem of fragmented internal knowledge. Currently, crucial documents are stored across multiple sources and disconnected platforms, leading to employees spending a significant amount of time searching for answers. Consequently, this project is established to query internal knowledge repositories from multiple sources using natural language directly within Slack, receiving instant, accurate answers while serving internal employees securely through a channel-level whitelist mechanism.
DESIRED OUTCOME	- Multi-Source Knowledge Retrieval: Delivery of a secure, Slack-integrated conversational assistant that enables internal employees to instantly query and retrieve information from multiple documentation sources (including Confluence, with extensible support for GitHub and internal documentation sites). - Channel-Level Whitelist Access Control: Successful implementation of a straightforward channel whitelist, ensuring only authorized Slack channels can interact with the Slack app bot without requiring complex, granular departmental permission controls. - Secure Corporate Deployment: Operation within a private corporate workspace leveraging enterprise-managed Slack SSO authentication and subscription features for seamless security management.
ACTION TO COMPLETION	- Multi-Source Ingestion and Indexing: Develop an automated pipeline to collect, chunk, and index internal documentation from multiple sources into a vector database. - RAG Engine Implementation: Set up the Retrieval-Augmented Generation core to process natural language queries and retrieve precise answers. - Slack App Bot Integration and Whitelist Configuration: Build the official conversational interface as a Slack app bot and configure channel-level whitelist restrictions to limit usage to designated channels within the private corporate workspace. - System Testing and Validation: Perform end-to-end testing of document retrieval accuracy, response latency, and channel security boundaries

BENEFITS OF PROJECT	- Eliminates time wasted searching across fragmented documentation by providing instant, accurate answers directly within the primary team communication tool via a dedicated Slack app bot. - Delivers immediate value through multi-source document support while establishing a robust, extensible foundation for future integration - Ensures high enterprise data privacy by operating within a private corporate workspace, utilizing enterprise Slack SSO, and enforcing simple channel whitelist restrictions.
PROJECTED SCHEDULE	- Research, Architecture Design and Source Ingestion: Define system requirements, design the RAG architecture, and plan the multi-source data extraction and vector indexing pipeline. - Core RAG and Channel Whitelist Planning: Map out the Large-Language Model retrieval logic and design the Slack channel whitelist security framework within a private workspace model. - Integration Blueprint and Testing Plan: Formulate the Slack app bot deployment strategy, evaluation metrics for response accuracy, and expansion plans.
PROJECTED BUDGET	Zero financial cost at the planning stage, utilizing open-source libraries, local development environments, and free-tier developer tools for Slack and Confluence integration.
PROJECTED TEAM AND RESOURCE REQUIREMENTS	- Student development team responsible for system architecture, RAG implementation, pipeline design, and documentation. - Development Environment: Python, Vector Database frameworks, and RAG libraries. - API and Integration Tools: Confluence API documentation, Slack Bot API, and developer workspace tools. - Academic References: Technical literature on Retrieval-Augmented Generation, access control mechanisms, and enterprise knowledge management systems.